

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/02

Paper 2 Core Paper

2 September 2013

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

Circle the question number of the question attempted

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
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7	
Section B	
8 / 9	
Total	

This document consists of **22** printed pages and **2** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 In translation, mRNA that is produced in transcription is used as the template for the synthesis of a polypeptide chain which will be further coiled and folded to form a functional protein. Fig. 1.1 shows a stage during translation.

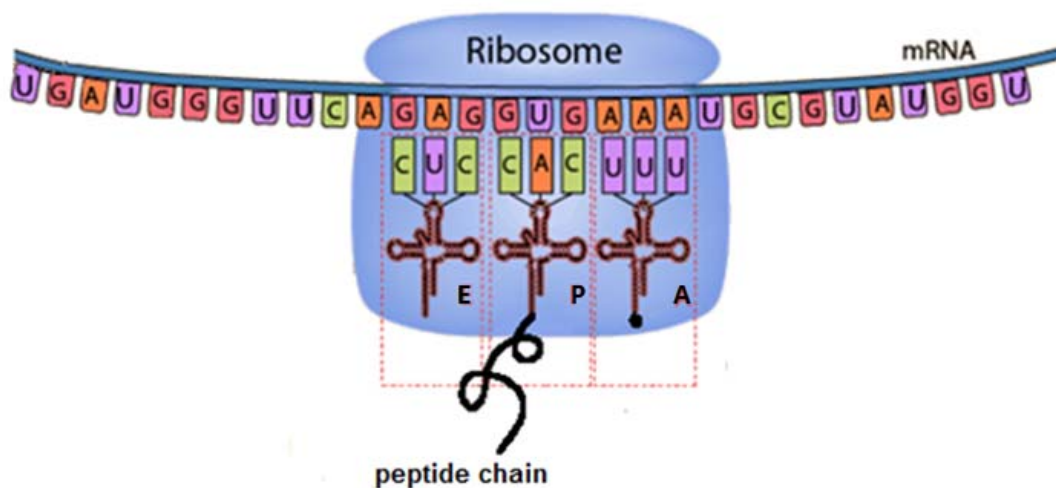


Fig. 1.1

- (a) Name the stage of translation shown in Fig. 1.1. [1]

- **Elongation;;**

- (b) State the number of amino acids found on the peptide chain. [1]

- **5;;**

- (c) During translation, Enzyme X was used to catalyse the reaction shown in Fig. 1.2.

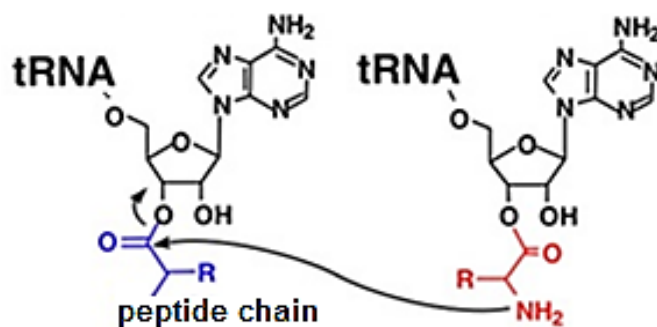


Fig. 1.2

- (i) State the name of Enzyme X. [1]

1. **Peptidyl transferase;;**

(ii) Using the information provided in Fig. 1.1 and Fig. 1.2, describe the process catalysed by Enzyme X. [3]

1. **peptidyl transferase catalyses the formation of a peptide bond;**
2. **The peptide chain/polypeptide bound to the tRNA in the P site separates from the tRNA;**
3. **and attaches (by a peptide bond) to the amino acid carried by the tRNA/aminoacyl tRNA in the A site;**
4. **The (peptide bond) is formed between the amino end of the amino acid in the A site;**
5. **and the carboxyl end of the peptide/growing chain in the P site;**
6. **one more amino acid is added to the polypeptide chain;**
7. **transferring the peptide/growing chain to the tRNA in the A site;**

(iii) Enzyme X is able to act on all the aminoacyl-tRNA found in the body. Describe the model by which the aminoacyl-tRNA binds to the enzyme. [2]

1. **According to the induced fit hypothesis/model;**
2. **The initial shape of the active site of enzyme may not be exactly complementary to that of the substrate;**
3. **However, as the substrate enters (and binds) to the active site, it induces a conformational change of the enzyme;**
4. **Such that the substrate can fit even more snugly into the active site;**
5. **Allowing enhanced interaction between the chemical groups of the substrate and the amino acid residues at active site for successful reaction;**

The rate of reaction of Enzyme X is affected by pH within the cytosol where translation is taking place. Fig. 1.3 was the result of an experiment to investigate how pH affected the rate of translation in normal cells.

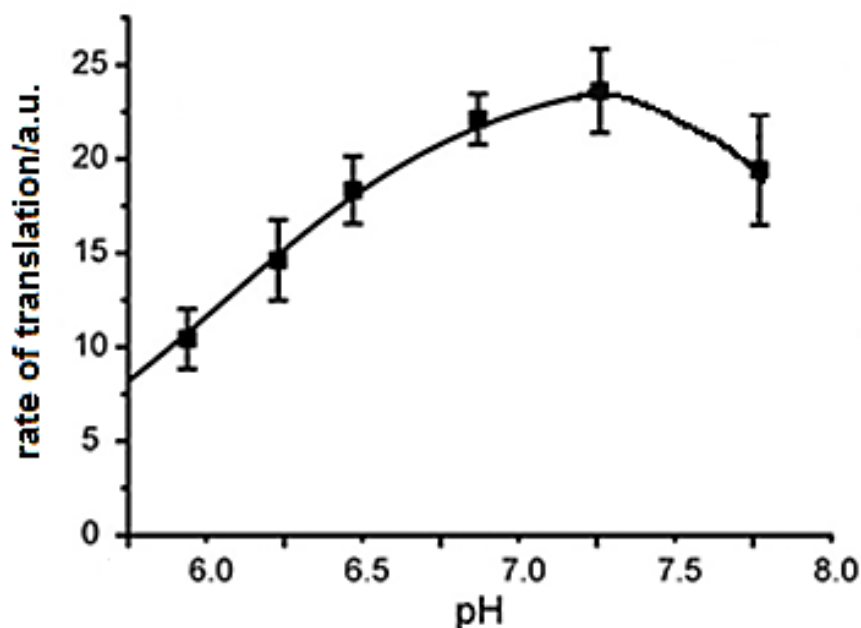


Fig. 1.3

(d) With reference to Fig. 1.3, describe how pH affects the rate of enzyme reaction. [2]

1. The rate of translation is at maximum of 22.5 a.u. at optimum pH of pH 7.25;;
2. At pH other than pH 7.25, the rate of translation decreases, reaching 8 a.u. at pH 5.75 and 20 a.u. at pH 7.75;;

(e) A toxin introduced into normal cells changes the pH of the cells, leading to a reduced rate of translation and hence bringing about cell death.

Suggest what the pH of the cells were after the introduction of the toxin and explain why protein synthesis was affected. [2]

1. pH 5.75 - pH 6.75 or pH 7.75; (state a value)
2. low pH alters the ionic charge of the acidic and basic R-groups of amino acids (of peptidyl transferase);
3. results in breakage of ionic bonds (and hydrogen bonds) between R groups of amino acids;
4. loss of 3-D conformation of (active site of) peptidyl transferase;
5. peptidyl transferase is denatured (and affects protein synthesis);
6. substrate/aminoacyl tRNA can no longer bind to active site of enzyme to form enzyme-substrate complex, decreased rate of translation;

[Total: 12]

- 2 Fig. 2.1 shows a cell that has undergone semi-conservative replication and is in the process of undergoing telophase and cytokinesis.

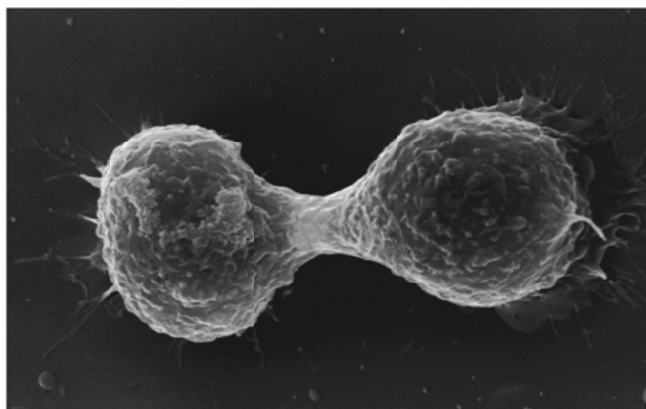


Fig. 2.1

(a) (i) State precisely when DNA replication occurs in the cell cycle. [1]

- **S phase of interphase ;**

(ii) State what is meant by semi-conservation replication. [2]

1. **The two parental DNA strands unwinds/separate into 2 strands;**
2. **by the breaking of hydrogen bonds between complementary bases;**
3. **and each parental strand act as a template for the synthesis of a new complementary strand;**
4. **2 new DNA molecules formed, each consisting of one original strand and one newly synthesized strand of DNA;**

(iii) Describe the roles of enzymes used in the process. [3]

1. **Helicase;**
2. **unwinding of DNA double helix (at origins of replication) by breaking of hydrogen bonds between complementary base pairs, (allowing separation of the two strands);**
OR
3. **Primase;**
4. **synthesizes RNA primer to initiate replication process;**
OR
5. **DNA polymerase;**
6. **catalyses the formation of phosphodiester bonds between adjacent nucleotides (of growing strand, synthesising daughter strand in 5' to 3' direction);**
OR
7. **another DNA polymerase;**
8. **removed RNA primers and replaced with DNA nucleotides;**
OR
9. **DNA ligase;**
10. **anneal sugar phosphate backbone / seals nicks between (Okazaki) fragments via formation of phosphodiester bonds to form continuous strands;**

(b) Describe the events that occur during the stage shown in Fig. 2.1. [3]

1. **Daughter chromosomes reach opposite poles of the cell;**
2. **Chromosomes uncoil, (lengthen and become indistinct) to form chromatin;**
3. **Spindle fibres disintegrate;**
4. **Nuclear envelope reforms (around the chromatin in each daughter cell);**
5. **Nuclei formed;**
6. **Nucleoli reappear (in each daughter nucleus);**
7. **cleavage furrow begins to form / Cell starts to divide by furrowing;**
8. **Constriction/contraction of ring of microfilaments in middle/center of the cell for outside inward**

An enzyme called telomerase is produced by some cells in the human body. Telomerase adds nucleotides to the ends of chromosomes.

Table 2.1 shows the telomerase activity and telomere length of some types of cell after cell division.

Table 2.1

type of cell	number of cell divisions	telomerase activity	telomere length
bone cell	limited	absent	shortened
cancer cell	unlimited	high	maintained
embryonic stem cell	unlimited	high	maintained
epidermis cell	limited	absent	shortened
liver cell	partially limited	low	slightly shortened
neurone	limited	absent	shortened

(c) Using the information in Table 2.1,

(i) State the consequences of high telomerase activity in a cell. [1]

1. **telomere length is maintained;**
2. **cell is able to continue to divide indefinitely/unlimited number of cell division;**
3. **cell might become cancerous;**

(ii) Explain the relationship between high levels of telomerase expression and cancer. [2]

1. **telomerase expression is a step in the multi-step process of cancer (where more than one somatic mutation is involved);**
2. **telomerase adds specific DNA sequence repeats to telomeres;**
3. **preventing the shortening of chromosomes to the critical length (beyond which cell division can no longer occur);**
4. **apoptosis is not triggered / cells with high levels of telomerase do not undergo apoptosis/replicative cell senescence;**
5. **cells continue to divide infinitely/uncontrollably and do not undergo apoptosis;**
6. **contributing to the development of cancer;**

[Total: 12]

- 3 During the first phase of the cell cycle, G1, there is a crucial cell-cycle checkpoint that a cell must pass to begin the process that leads to cell division. If changes in the structure of DNA are detected during G1, the cell cycle is halted until DNA is repaired.

p53 gene located on chromosome 17 codes for p53 and is known as a tumour suppressor gene. Mutations of one of the two *p53* gene cause a hereditary disorder known as Li-Fraumeni syndrome.

Fig. 3.1 is a simplified flow chart showing the sequence of events that occurs after a change in the structure of DNA is detected.

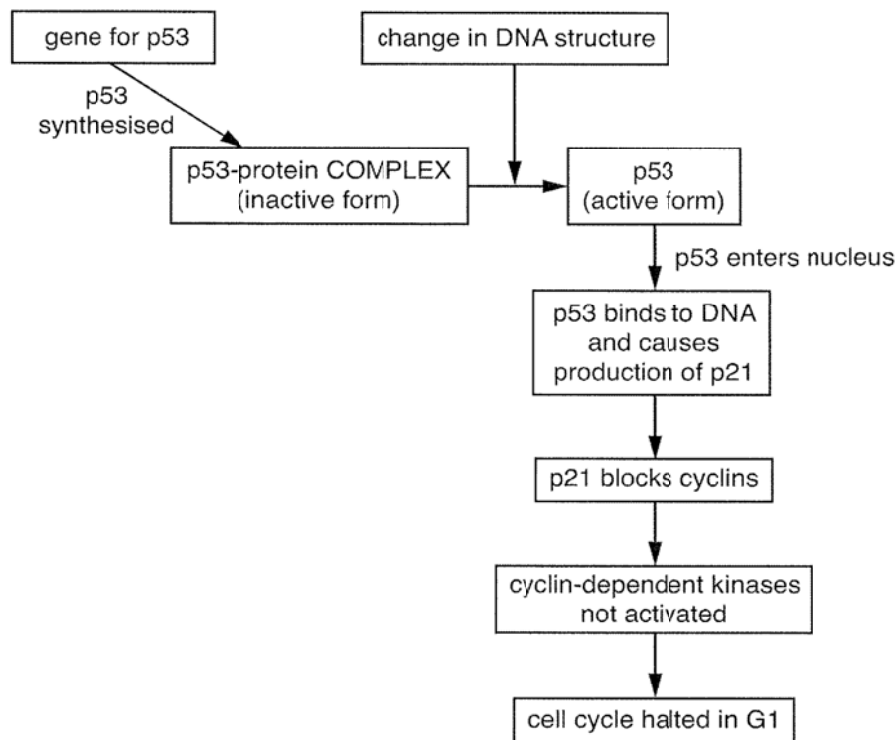


Fig. 3.1

(a) What is a tumour suppressor gene? [1]

1. **encodes protein that prevent uncontrolled cell division;;**

In the inactive form, p53 forms a p53-protein complex. In the active form, p53 is no longer attached to the protein.

(b) Suggest why only the active form of p53 can enter the nucleus. [1]

1. **active p53 is not complexed with/joined to the protein;**
 2. **small enough to pass through nuclear pore;**
(Credit: reverse argument for inactive form)
- OR
3. **complementary shape;**
 4. **to the binding site of the carrier / channel protein;**

(c) Using the information in Fig. 3.1, explain how mutation of *p53* gene could lead to cells continuing to divide even after a change in the structure of DNA is detected. [3]

1. non functional p53 synthesized;
2. unable to recognise and bind to DNA;
3. no p21 synthesized;
4. cyclins not blocked/can activate cyclin-dependent kinases;
5. cell cycle not halted/goes beyond G1;
6. progresses into S phase of interphase;
7. DNA replication continues, cell continues into cellular division;

The effectiveness of anti-cancer drugs against metastatic colorectal cancer may be determined by growing human tumours in culture. The effectiveness of two drugs, T138067 and vinblastine, against the tumours was assessed. The two drugs were added to the tumours in culture on days 5, 12 and 19. The volumes of the tumours were compared with the volumes of tumours that were not treated with any drugs.

The results are shown in Fig. 3.2.

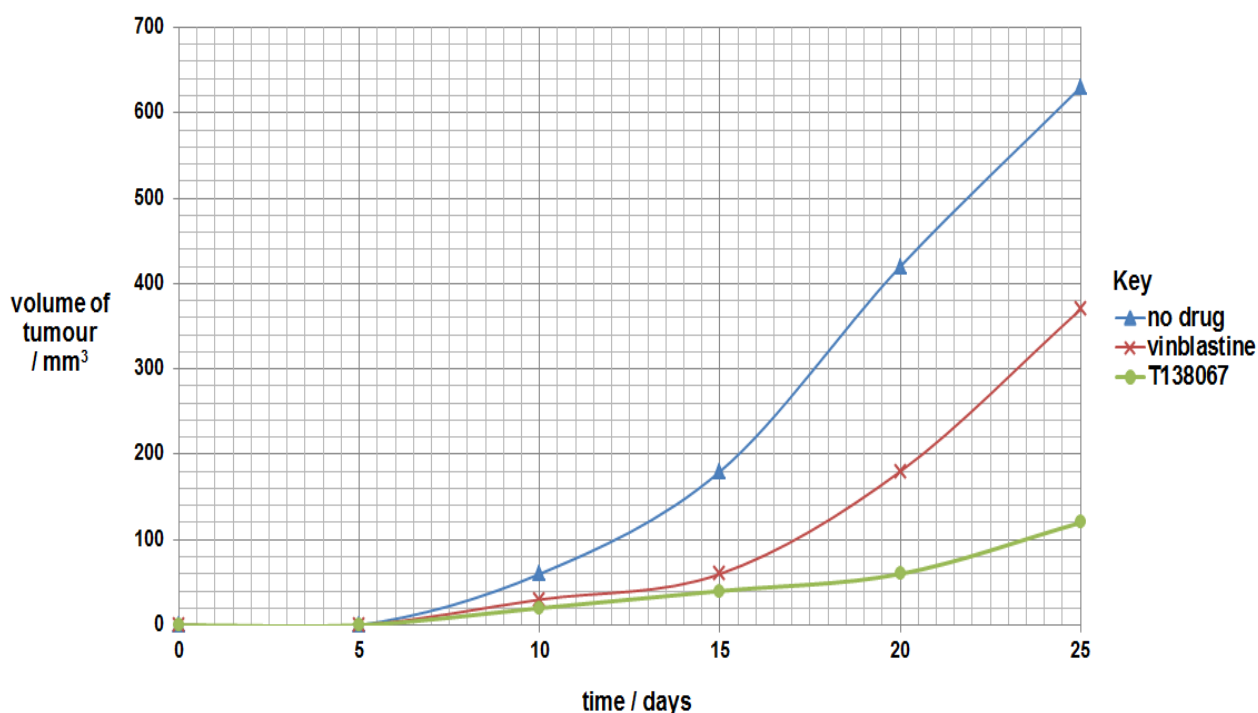


Fig. 3.2

(d) With reference to Fig. 3.2,

(i) conclude the effectiveness of T138067 on tumours. [1]

1. T138067 is effective in treating of tumours / not completely effective in stopping tumour growth/tumours continue to grow;
2. At 25th day, the volume of tumour treated with T138067 is much smaller at 120mm³ compared to volume of untreated tumour at 630(±10)mm³;
3. Difference of 510(±10)mm³ between tumour treated with T138067 compared to untreated tumour / 5.25X smaller;

(ii) compare the effectiveness of the two drugs when used to treat patients suffering from metastatic colorectal cancer. [2]

1. T138067 is a more effective drug than vinblastine / vinblastine is a less effective drug against tumours;
2. At 25th day, volume of tumour treated with T138067 is 120mm³ while volume of tumour treated with vinblastine is 370mm³;
3. Tumour treated with T138067 is 3.1X smaller;
4. Tumour treated with T138067 grew at a slower rate of 4.8mm³ while that treated with vinblastine grew at a faster rate of 14.8mm³;
OR
5. Compared to tumour that is untreated, T138067 slowed down growth by 510(±10)mm³/41.3% while vinblastine slowed down growth by 260(±10)mm³/81.0%;
6. T138067 is more effective than vinblastine by 1.96X;
7. Up to the 10th/15th day, little difference/similar effectiveness in effectiveness between vinblastine and T138067 against tumour;
8. Difference of only 10-20mm³;
(Credit: any other valid data analysis)

Vinblastine disrupts the formation of the spindle apparatus

(e) Explain how vinblastine acts as an anti-cancer drug. [2]

1. growth of tumour/cell division involves mitosis;
2. spindle apparatus required for stages of mitosis / use of vinblastine results in absence of spindle apparatus;
3. no attachment of spindle apparatus to attach to centromeres of chromosomes during prophase;
4. Chromosomes will not align along metaphase plate during metaphase;
5. Centromeres will not divide and chromatids will not separate during anaphase;
6. mitosis stops / metaphase to anaphase to telophase, cannot proceed / cell division stops / prevents uncontrolled cell division;
7. preventing tumour cells from undergoing further mitosis/cell division;
(pt 3-5: max 1)

[Total: 10]

4 Fig. 4.1 shows a Human Immunodeficiency Virus.

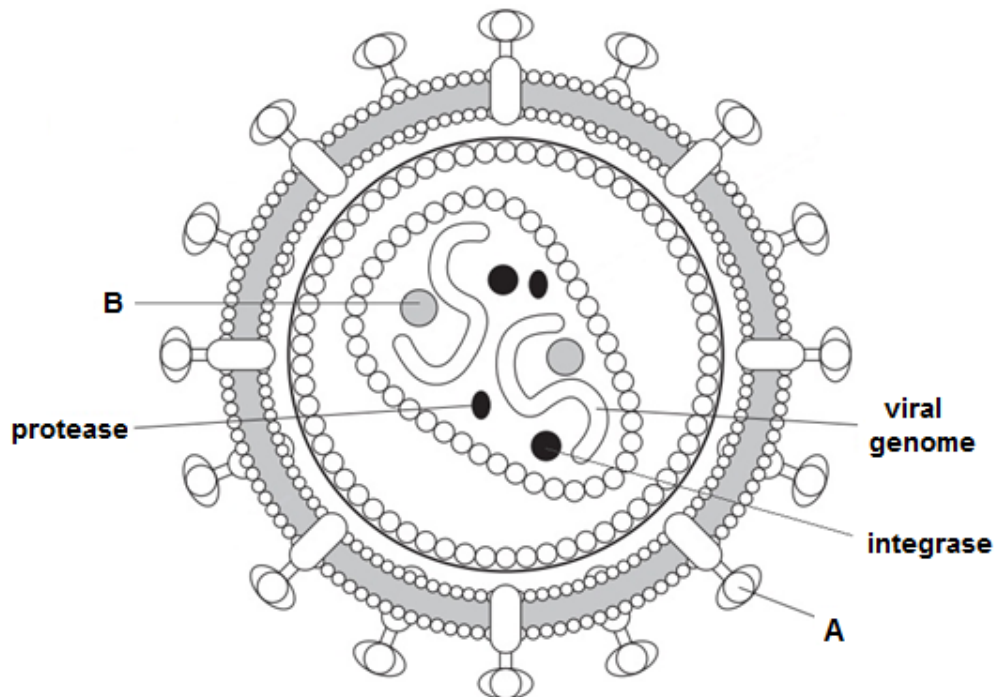


Fig. 4.1

(a) Name the structures A and B in Fig. 4.1. [1]

1. **A – glycoprotein / gp120;**
2. **B – reverse transcriptase;**

(b) Describe how the viral genome is released into host cell after a conformational change has occurred in gp41. [2]

1. **Fusion of viral envelope with the host cell's plasma membrane;**
2. **allows the nucleocapsid to enter the target cell;**
3. **Uncoating occurs;**
4. **Capsid is removed by cellular enzymes;**

(c) Fig. 4.2 shows a generalized graph of the relationship between HIV RNA copies and T lymphocytes counts over the average course of untreated HIV infection.

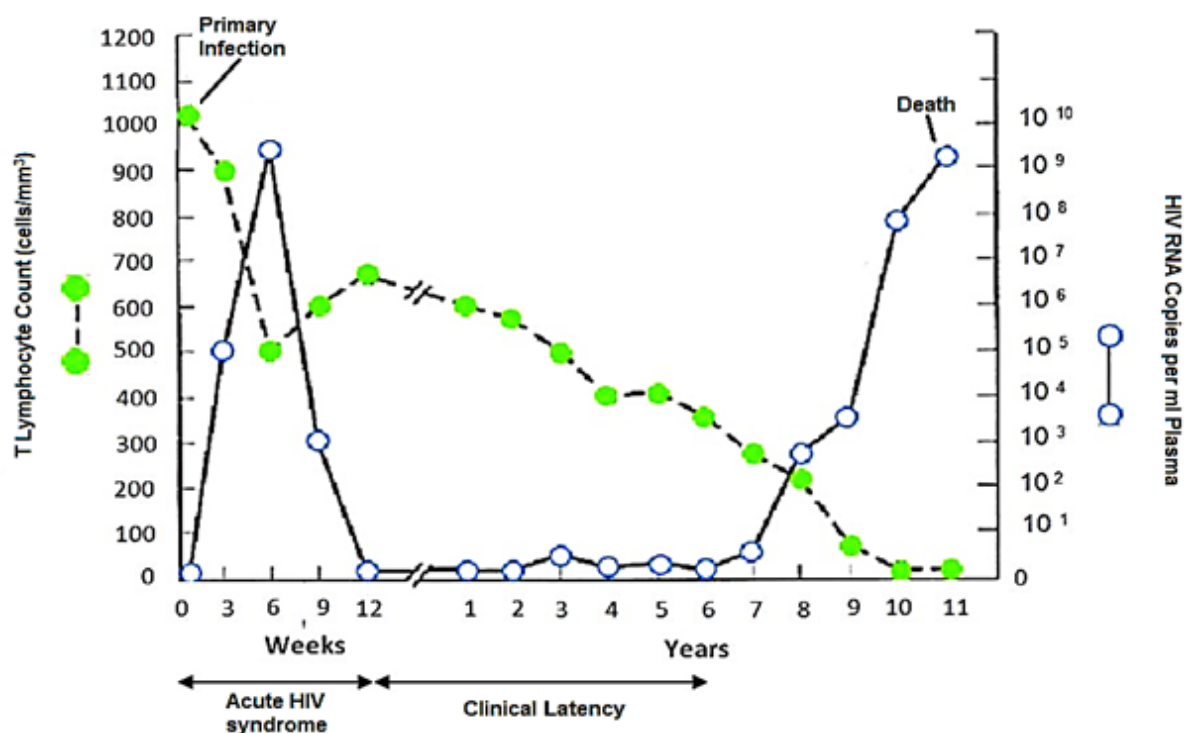


Fig. 4.2

(i) With reference to Fig. 4.2, describe and explain what happens during the period of clinical latency. [3]

1. From the 12th week to 6th year (since primary infection), the HIV RNA copies remain constant at 3/ less than 10/~ 0 copies per ml plasma;
2. Replication occurs;
3. Reverse transcriptase catalyses the synthesis of a complementary DNA strand (cDNA) from viral RNA;
4. synthesis of a second DNA strand which is complementary to cDNA;
5. double stranded DNA is integrated/incorporated into the T lymphocyte's DNA/genome (as a provirus) by integrase;
6. number of free infectious virus is low;
7. number of HIV RNA copies is low;

(ii) Explain how HIV infection causes death in the 11th year from primary infection. [2]

1. Replication of HIV destroys T lymphocytes;
2. resulting in low T lymphocytes counts;
3. the immune system weakens and develops AIDS;
4. susceptible to opportunistic infections which can result in death;

- (d) The management of AIDS normally includes the use of multiple anti-retroviral drugs that act on different stages of the HIV life cycle in an attempt to control HIV infection. One such drug is alpha interferon which inhibits an enzyme necessary in the maturation and release stage of the HIV life cycle.

Identify the enzyme and explain how HIV infection is controlled. [2]

1. **HIV protease;**
2. **HIV polyprotein cannot be cleaved (into individual functional proteins);**
3. **like capsid proteins, reverse transcriptase and integrase (any 2);**
4. **virus remained immaturred / non-infectious / non-functional;**
5. **unable to infect other cells/ unable to carry out further infection;**

[Total: 10]

5 A plant breeder carried out three crosses as follows:

- **cross 1** - pure breeding tomato plants with uniformly pigmented red-coloured fruit were crossed with pure breeding tomato plants with dark patches at the base of the orange-coloured fruit
- **cross 2** - the offspring of cross 1 (offspring 1) were crossed with each other
- **cross 3** - offspring 1 were crossed with pure breeding tomato plants with dark patches at the base of the orange-coloured fruit

The number of offspring in each phenotype was counted.
Table 5.1 shows the results of the three crosses.

Table 5.1

cross	parental phenotype		total offspring	number of offspring of each phenotype
1	pure breeding plants with red uniformly pigmented fruit	X pure breeding plants with orange fruit with dark patches	150	150 red uniformly pigmented fruit
2	Offspring 1 plants with red uniformly pigmented fruit	X Offspring 1 plants with red uniformly pigmented fruit	172	24 red fruit with dark patches 98 red uniformly pigmented fruit 46 orange fruit with dark patches 4 orange uniformly pigmented fruit
3	Offspring 1 plants with red uniformly pigmented fruit	X pure breeding plants with orange fruit with dark patches	210	23 red fruit with dark patches 83 red uniformly pigmented fruit 85 orange fruit with dark patches 19 orange uniformly pigmented fruit

(a) Using the symbols **R** for red fruit and **r** for orange fruit, **D** for uniform pigment and **d** for dark patches, draw a genetic diagram to explain the results in Cross 3. [4]

parental phenotypes: red uniformly pigmented fruit X orange fruit with dark patches

parental genotypes: $\frac{RD}{rd}$ X $\frac{rd}{rd}$;;

Gametes $\frac{RD}{rd}$ $\frac{Rd}{rd}$ $\frac{rD}{rd}$;;

Punnett Square

	$\frac{RD}{rd}$	$\frac{Rd}{rd}$	$\frac{rD}{rd}$	$\frac{rd}{rd}$
$\frac{rd}{rd}$	$\frac{RD}{rd}$	$\frac{Rd}{rd}$	$\frac{rD}{rd}$	$\frac{rd}{rd}$

offspring genotype : $\frac{RD}{rd}$: $\frac{Rd}{rd}$: $\frac{rD}{rd}$: $\frac{rd}{rd}$;;

offspring phenotype : red uniformly pigmented fruit : red fruit with dark patches : orange uniformly pigmented fruit : orange fruit with dark patches ;

offspring phenotypic ratio : 83 : 23 : 19 : 85 ;

(b) Explain the deviation of the observed results from the expected dihybrid ratio. [3]

1. Expected dihybrid ratio is 1:1:1:1 while observed test cross results is approx. 4:1:1:4 which deviates from the expected ratio;
2. The 2 genes for colour of fruit and pigmentation are linked;
3. which means they form a linkage group and are usually pass into the same gamete and are inherited together /do not undergo independent assortment and segregation during meiosis;
4. crossing over between the 2 linked genes on non-sister chromatids of homologous chromosomes during prophase 1 of meiosis 1 occur ;
5. to form recombinant gametes – Rd and rD;
6. since crossing over is a chance event, there are lower chances of getting recombinant gametes;
7. thus the observed offspring of test cross are majority of parental combinations and minority of recombinants combinations;

Observations showed that in tomato plants, there are 3 different types of pigmentation at the base of the fruit. Pigmentation in these fruits is controlled by 2 proteins involved in the same metabolic pathway for production of pigments, Inhibitor E and Enzyme B. Uniform pigmentation is produced in the presence of two copies of 'e' alleles and one copy of 'B' alleles. Dark patches are produced in the presence of two copies of 'e' alleles. No pigmentation is produced in the presence of at least one copy of 'E' allele.

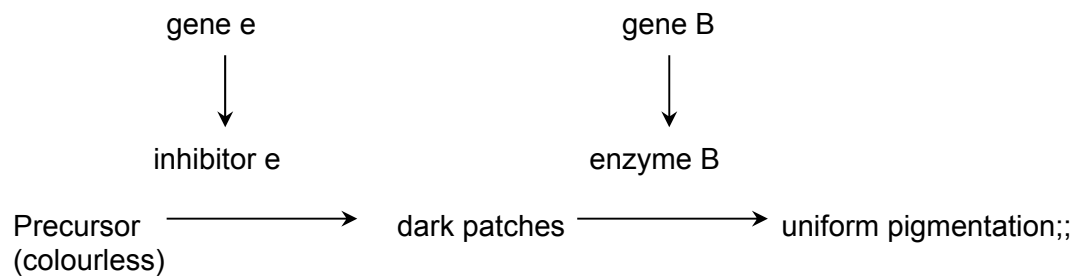
(c) Suggest a possible mode of inheritance for pigmentation at the base of the fruit. [1]

- **Dominant epistasis;;**

(d) Explain how the interaction between the two genes *E* and *B* gives rise to pigmentation. [3]

1. The pigmentation is controlled by a pair of genes (gene *E/e* and gene *B/b*) occupying different loci;
2. the epistatic gene determines the production of pigments;
3. the dominant allele (*E_*) results in no pigmentation;
4. homozygous for the recessive allele (*ee*) results in uniform pigmentation or dark patches, depending on the genotype of the gene *B/b*;
5. the hypostatic gene determines the type/intensity of pigmentation;
6. the dominant allele (*B_*) determines uniform pigmentation;
7. homozygous for the recessive allele (*bb*) determines dark patches;
8. In the interaction between these two genes, the presence of allele *E* hides the effects of either *B_* or *bb*, producing no pigmentation;
9. *E_* is epistatic to (any genotype of the) *B/b* gene;

(e) Explain the biochemical basis for this form of epistasis. [2]



Award 1 mark for correct genes and gene products;;

[Total: 13]

- 6 Fig. 6.1 shows an electron micrograph of a double membranous organelle, mitochondrion.



Fig. 6.1

(a) Describe the structure of membrane. [5]

1. Made up of a phospholipid bilayer;
2. The thickness is about 7nm;
3. The (hydrophilic) phosphate heads of the phospholipids face outwards into the aqueous environment/water (inside and outside the cell);
4. The (hydrophobic) hydrocarbon chains/ fatty acid tails face inwards and create a hydrophobic interior (core)/face away from aqueous medium;
5. Hydrophobic fatty acid tails are buried inside the two hydrophilic layers formed by the phosphate groups;
6. Phospholipids are held by (weak) hydrophobic interactions;
7. Most protein molecules embedded in/float about in the phospholipid bilayer;
8. In an irregular pattern/randomly/forming a fluid mosaic pattern;
9. Peripheral proteins (loosely) bind to the surface of the membrane / attached regions of the integral proteins;
10. Integral proteins penetrate/span all the way through the membrane;
11. Glycoproteins and glycolipids have (short branching) carbohydrate chains;
12. Contain cholesterol;
13. Membranes are asymmetric/the two sides of a membrane can differ in composition and function;

(b) From your biological knowledge, explain the adaptation of the mitochondrion's double membrane for its role in the production of energy. [2]

1. Membranes impermeable to H^+ /protons / contain proton pumps;
2. to create proton reservoir/gradient;;
- OR
3. Highly folded inner membrane to increase surface area;
4. For embedding of ATP synthase / electron carriers;
5. Movements of protons through the stalked particles release energy for the synthesis of ATP
- OR
6. Compartmentalisation/separation within the mitochondrion;
7. so that reactions may occur in different areas;

Fig. 6.2 shows a stage in respiration that takes place during the oxidation of glucose.

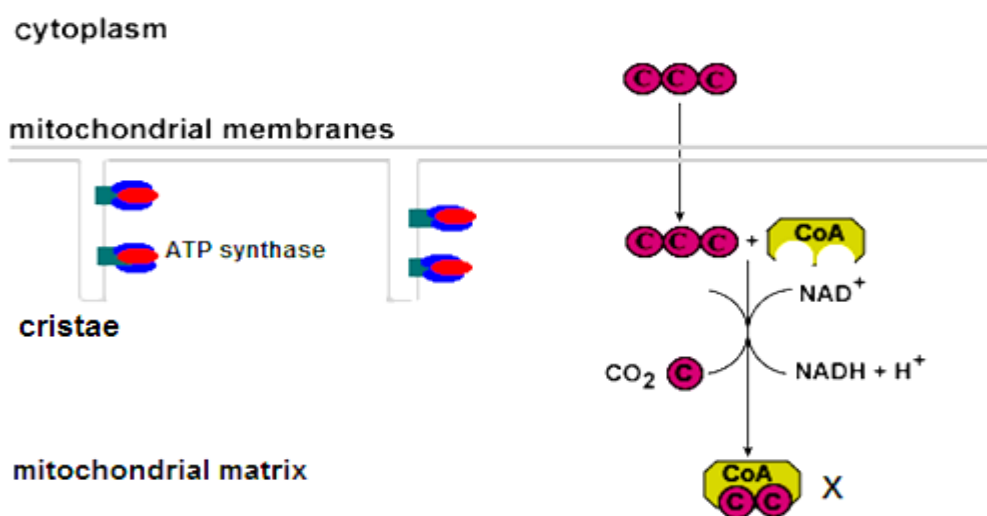


Fig. 6.2

(c) Explain what happens to Molecule X. [3]

1. Molecule X is acetyl CoA
 2. Acetyl CoA and oxaloacetate are condensed to form citrate;
 3. Oxaloacetate displaces the coenzyme A;
 4. Citrate undergoes oxidative decarboxylation to form α -ketoglutarate;
 5. A molecule of CO_2 is lost, (and the remaining 5C compound is oxidised by NAD^+)
 6. α -ketoglutarate undergoes oxidative decarboxylation to form succinate;
 7. A molecule of CO_2 is lost, (and the remaining 4C compound is oxidised by NAD^+);
 8. Succinate undergoes oxidation to form fumarate;
 9. Fumarate is converted to form malate;
 10. Malate undergoes oxidation to regenerate oxaloacetate;
- Pt 4 and 6: award once

(d) List two ways in which photophosphorylation differs from oxidative phosphorylation.
[3]

Features	No.	Photophosphorylation	No.	Oxidative phosphorylation
Process	1a	In photosynthesis	1b	In aerobic respiration
Location	2a	Thylakoid membrane of chloroplast	2b	Inner membrane of mitochondrion
Involvement of light energy	3a	Light energy is required for photolysis / required to energise electrons in special chl a	3b	Light energy is not required
Source of energy for ATP synthesis	4a	Energy for synthesis of ATP comes from light	4b	Energy for synthesis of ATP comes from the oxidation of glucose
Electron donors	5a	Water is the electron donor in non-cyclic reaction	5c	NADH and FADH ₂ are electron donors
	5b	Photosystem I is the electron donor in the cyclic reaction		
Electron acceptors	6a	NADP ⁺ is the final electron acceptor in the non-cyclic reaction	6c	Oxygen is the final electron acceptor
	6b	PSI is the electron acceptor in the cyclic reaction		

(Credit: any other valid comparisons)

[Total: 12]

- 7 The takahē, *Porphyrio hochstetteri*, is a flightless bird that is restricted to a small area of the South Island of New Zealand. It is one of only two remaining species of large, flightless, herbivorous birds from New Zealand. The other remaining species is found on the North Island of New Zealand. Their flighted ancestor came over from Australia millions of years ago.

The takahē was thought to be extinct, but a small population with low genetic variation was discovered in 1948 among the remote mountains of the South Island. The takahē is a grassland specialist and lives in alpine grassland dominated by tussock grasses as shown in Fig. 7.1.



Fig. 7.1

- (a) Explain why it has been possible for flightless birds, such as the takahē, to evolve in New Zealand. [3]
1. Flighted birds migrated to New Zealand;
 2. (genetic) variation in the takahē's ability to fly due to (spontaneous) mutation;
 3. Some birds were flightless while some were flighted;
 4. Selection pressure – absence of mammals, abundant food / resources on ground / nesting sites available on the ground / few or no land predators;
 5. As flight requires considerable energy;
 6. Flightless birds with higher fitness/are better adapted to the environment will be at a selective advantage;
 7. They will survive to maturity, mate / reproduce and pass down favourable alleles to their offspring;
 8. Over time, change in allele frequency occurs, leading to evolution (and flightless birds predominates);

(b) Differences in the DNA sequences that encode cytochrome b in the two species of flightless birds from New Zealand have been measured.

(i) When differences in cytochrome b gene is plotted against time since colonisation, the plot of the line is straight. Describe how this supports the neutral theory of molecular evolution. [2]

1. **Plot of line being straight indicates that the rate of mutation is constant / steady;**
2. **Mutations are silent and there are small number of changes;**
3. **There is no effect on the phenotype i.e. cytochrome b of the organisms;**
4. **No selective advantage or disadvantage is involved;**

(ii) Suggest why the cytochrome b gene is used to measure changes in DNA sequences between the two species of flightless birds. [2]

1. **The cytochrome b gene was present in both species and is therefore a good basis for comparison;;**
2. **Changes in gene occur very slowly (as mutations are silent);;**
3. **Changes in the gene is restricted as unfavourable mutations will result in gene being eliminated from gene pool;;**

(c) The takahē is similar in appearance to another flightless bird found on the Desventuradas Islands off the coast of Chile even though molecular studies have shown that they do not share a common ancestor. Explain how biogeography support the evolutionary deductions based on molecular homology. [2]

1. **Biogeography is the study of the geographical distribution of species;**
2. **Unrelated species in distinct region may look similar to each other/different species evolved and have common features;**
3. **due to convergent evolution;**
4. **Unrelated species faced similar selection pressures in their environment;**
5. **due to similar conditions on the different islands;**

Another method to determine evolutionary relationships of different animals would be based on anatomical homology.

Fig. 7.2 shows the relationship between some animals which was determined by comparing the arrangement of the bones in their forelimbs, which had a basic pentadactyl structure.

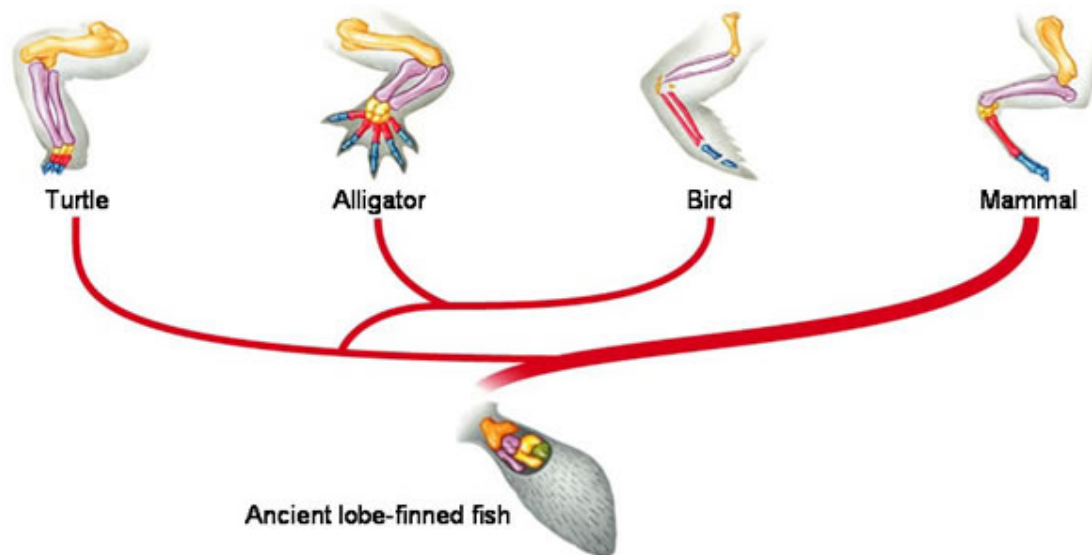


Fig. 7.2

(d) Explain how the anatomical homology in Fig. 7.2 provides evidence to support the theory of evolution. [2]

1. **Although the forelimbs had different functions;**
2. **they basically have the same arrangement/similar structure;**
3. **which is a basic pentadactyl/ five digit forelimb structure;**
4. **Therefore they were likely to have originated/derived from a common ancestor;**
5. **Which is the ancient lobe-finned fish;**
6. **and had been modified over time/ descent with modification;**

[Total: 11]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Describe the main structural features and function(s) of collagen and glycogen. [8]
- (b) Explain how a change in the nucleotide sequence of DNA leads to cystic fibrosis. [8]
- (c) Explain how the recessive allele that results in cystic fibrosis may be preserved in a natural population. [4]

[Total: 20]

- 9 (a) Describe the main structural features of the *lac* operon and explain how the presence of lactose in the growth medium affects the *lac* operon. [6]
- (b) Describe the role of insulin and glucagon in regulating blood glucose concentration. [8]
- (c) Explain how a nerve impulse is passed across a cholinergic synapse. [6]

[Total: 20]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a)** Describe the main structural features and function(s) of collagen and glycogen. [8]

structural features of glycogen

- 1. A polysaccharide made from α glucose;**
- 2. A backbone of α -glucose residues held together by α -1, 4 glycosidic bonds;**
- 3. Highly branched with branches formed by α -1, 6 glycosidic bonds;**
- 4. angle of these bonds causes the chain to coil helically into a more compact shape;**
- 5. Most of the -OH groups are projected into the interior of the helix;**

functions of glycogen

- 6. food and energy stores in animals / compact nature of glycogen allows packing of many monomers per unit volume;**
- 7. provides a useful energy reserve for liver and muscles;**
- 8. Hydrolysis of glycogen in liver and muscle cells releases glucose when the demand for glucose increases. E.g. During exercise;**
- 9. can be hydrolysed by enzymes more efficiently / easily broken down into its glucose monomers (to be used as respiratory substrates);**
- 10. Large and insoluble nature of glycogen does not allow its diffusion out of the cell and do not change the osmotic potential within the cell;**

structural features of collagen

- 1. made up of amino acids monomers linked together by peptide bonds;**
- 2. Each (helical) polypeptide is in the shape of a loosely wound helix;**
- 3. high proportion of proline residues;**
- 4. almost every 3rd amino acid in each chain is glycine;**
- 5. glycine's small size allows the three strands of the polypeptide to (lie close together and) form a tight coil;**
- 6. The three polypeptides are wound around each other / twisted into / form a tropocollagen/triple helix;**
- 7. bound to one another by inter-molecular hydrogen bonds;**
- 8. tropocollagen runs parallel to each other to form fibrils and the ends of the molecule are staggered;**
- 9. Covalent cross-links form between (the carboxyl end of) one tropocollagen and (the amino end of) another tropocollagen;**
- 10. Fibrils unite to form fibres;**

function of collagen

- 11. structural protein found in connective tissue / in the basement membrane (in skin, tendons, cartilage, bones, teeth and the walls of blood vessels of humans and almost all animals);**

[Total: 8]

(b) Explain how a change in the nucleotide sequence of DNA leads to cystic fibrosis. [8]

- 1. Cystic fibrosis (CF) is an inherited disease where the mucus glands are affected.**
- 2. caused by a mutation of the cystic fibrosis transmembrane conductance regulator (CFTR) gene found on chromosome 7;**
- 3. a mutation results in deletion of 3 nucleotides CTT;**
- 4. malformed CFTR protein loses an amino acid, phenylalanine;**
- 5. at position 508**
- 6. CFTR protein is a transmembrane chloride channel that allows efflux of chloride ions out of the epithelial cells;**
- 7. mutation of the CFTR gene results in a change in the 3D conformation/ configuration of the chloride channel;**
- 8. channel is no longer able to transport chloride ions;**
- 9. prevent normal efflux of chloride ions out of epithelial cells;**
- 10. chloride ions accumulate in the epithelial cells;**
- 11. attracting water into the epithelial cells causing thick and sticky mucus;**
- 12. cells that line the passageways of the lungs/pancreas/other organs produce thick and sticky mucus.**
- 13. mucus obstruct the airways**
- 14. leading to severe problems with breathing/ resulting in bacterial infections in the lungs/ permanent lung damage/ formation of scar tissue (fibrosis) and cysts in the lungs;**
- 15. mucus blocks the duct / interferes with the function of the pancreas;**
- 16. preventing enzymes and hormones from reaching the intestines to aid digestion and assimilation, leading to digestive problems;**
- 17. reduces life expectancy;**

(c) Explain how the recessive allele that results in cystic fibrosis may be preserved in a natural population. [4]

- 1. Diploidy ;**
- 2. individuals have 2 sets of chromosomes;**
- 3. The diploid nature of (most) eukaryotes hides a considerable amount of genetic variation from selection;**
- 4. Recessive allele that results in cystic fibrosis is less favourable / harmful in the present environment than its dominant counterparts/the dominant allele hence the recessive allele can persist/exist in the heterozygotes individuals in a population;;**
- 5. Only alleles present in the homozygous recessive organism will be expressed in the phenotype;**
- 6. and be exposed to (environmental) selection / elimination;**
- 7. Very few of the recessive alleles can be eliminated from the population in each generation;**
- 8. Known as heterozygote protection;**

[Total: 20]

- 9 (a) Describe the main structural features of the *lac* operon and explain how the presence of lactose in the growth medium affects the *lac* operon. [6]

1. An operon is a cluster of genes that make up a single transcription unit;
2. The *lac* operon encode the enzymes involved in lactose metabolism;

The *lac* operon consist of:

3. *lacP* (Promoter): binding site of RNA polymerase;
4. *lacO* (Operator): binding site of the *lac* repressor;
5. CAP Binding Site, binding site of CAP (catabolite activator protein);
6. *lacZ* codes for β -galactosidase;
7. which hydrolyses lactose into glucose and galactose;
8. *lacY* codes for *lac* permease;
9. which is a membrane protein for the transport of lactose into the cell;
10. *lacA* codes for β -galactoside transacetylase;
11. which is involved in the hydrolysis of lactose;

12. allolactose is formed when lactose enters the cell and is cleaved by β -galactosidase;
13. allolactose recognises and binds to the allosteric site of the *lac* repressor;
14. *lac* repressor changes its conformation from active to inactive;
15. *lac* repressor unable to bind to the operator;
16. RNA polymerase recognises and binds to promoter;
17. transcribe the genes of the *lac* operon/ transcription occurs;

(b) Describe the role of insulin and glucagon in regulating blood glucose concentration.

[8]

- 1. Blood glucose concentration increases above normal level of 90 mg per 100 cm³ of blood;**
- 2. Islets of Langerhans of the pancreas detect the deviation;**
- 3. β cells of islets of Langerhans secrete insulin;**
- 4. Insulin recognises and binds to receptor sites of receptors on the surface membrane of target cells;**
- 5. to stimulate glycogenesis, an increase in the rate of conversion of glucose to glycogen;**
- 6. for storage in liver cells and muscle cells;**
- 7. to stimulate an increase in the rate of respiration using glucose;**
- 8. as respiratory substrate which will be broken down and oxidised to form carbon dioxide and water;**
- 9. to stimulate an increase in the permeability of plasma membrane to glucose;**
- 10. to increase the rate of uptake of glucose from the blood by almost all body cells especially muscle cells;**
- 11. to stimulate an increase in the rate of conversion of excess glucose to fats;**
- 12. for storage in adipose tissues;**
- 13. Blood glucose concentration decreases and return back to normal level;**

[Max 6m]

- 14. Blood glucose concentration decreases below normal level of 90 mg per 100 cm³ of blood;**
- 15. Islets of Langerhans of the pancreas detect the deviation;**
- 16. α cells of islets of Langerhans secrete glucagon;**
- 17. Glucagon recognises and binds to receptor sites of receptors on the surface membrane of liver/target cells;**
- 18. To stimulate glycogenolysis, an increase in the rate of breakdown of glycogen to glucose;**
- 19. to stimulate gluconeogenesis, an increase in the rate of conversion of amino acids and glycerol to glucose;**
- 20. Blood glucose concentration increases and return back to normal level;**

(c) Explain how a nerve impulse is passed across a cholinergic synapse. [6]

1. Before the presynaptic membrane is depolarised, there are few Ca^{2+} ions inside the synaptic knob;
2. The arrival of an impulse/action potential at the synaptic knob causes voltage-gated Na^+ channels and voltage-gated Ca^{2+} channels to open;
3. resulting in influx of Na^+ ions and Ca^{2+} ions into the synaptic knob from the synaptic cleft;
4. The influx of Ca^{2+} ions causes vesicles containing neurotransmitters/ acetylcholine to move to the presynaptic membrane and fuse with it;
5. emptying the neurotransmitters into the synaptic cleft by exocytosis;
6. The Ca^{2+} ions are pumped out of the synaptic knob by active transport using ATP;
7. Acetylcholine diffuse across the 20nm synaptic cleft;
8. usually in less than 0.5 ms;
9. Acetylcholine is complementary in shape to specific binding sites of receptors;
10. and recognises and binds to the binding sites of receptors on the postsynaptic membrane;
11. This changes the shape of the receptor protein, opening chemically-gated Na^+ channels on the postsynaptic membrane;
12. Influx of Na^+ ions into the cytoplasm along concentration gradient depolarise the postsynaptic membrane;
13. The membrane potential becomes more positive than resting potential;
14. The local potential that develops is known as an excitatory postsynaptic potential (EPSP);
15. If the membrane potential of the postsynaptic membrane reaches threshold potential (-50mV), an action potential is generated in the postsynaptic neurone and travels down the axon to the next synapse;
16. The synaptic cleft contains an enzyme, acetylcholinesterase, which hydrolyses each acetylcholine molecule into acetate and choline. The choline is taken back into the presynaptic neurone to combine with acetyl coA to form acetylcholine;

[Total: 20]